

Technical Appendix — Markout Study

Number-to-source map, audit trail, and reproducibility

Quantitative Research — Internal

2026-07-05 · Version 6 · Internal Research

1. Number-to-source map

Every headline number maps to the script that produces it and the output it is read from.

Claim	Value	Script	Output
Universe	7,332,013 entries / 32,949 wallets	build_entries.py	out/entries/part_*.parquet
Cohort	1,694 wallets / 812,616 entries	cohort_K_{define,price}.py	out/cohort_K_entries.parquet
Effective sample	334 days / 179,402 wallet-days / 252,223 wallet-coin-days	verify_v4.py	out/v4.json
Cost schedule	BTC 8 / ETH 8 / SOL 10 / HYPE 14 bp round-trip	mkcommon.py (COST_BPS)	—
Term-structure table + CI	Appendix A	verify_v2.py	out/termstructure_ci.json, out/core_markout_table.md
Long/short raw vs neut	BTC 24h raw −23.4/+17.2, neut −0.4/−3.4	verify_v4.py	out/v4.json
Event study	pre-entry +18 to +33 bp	pi_compute.py	out/eventstudy.json
Conditional pooled slope	−0.058 [−0.108, −0.018], n=590,910	verify_v5.py	out/v5.json
Size within coin-month	no increase with size	verify_v4.py	out/v4.json
Static ρ (canonical)	Pearson +0.00 / Spearman +0.02, n=936	verify_v5.py	out/v5.json
Rolling decile / IC	top-field +4.0@4h / +10.3@8h; IC +0.093 [+0.061,+0.126]	cohort_M2_recurrence.py, verify_v4.py	out/v4.json
Recurrence train-only	≥ 2 121/88.7 · ≥ 3 27/8.3 · ≥ 4 5/0.5 (p<0.001)	verify_v5.py	out/v5.json
Recurrence full (exploratory)	≥ 2 213/185 · ≥ 3 72/35 · ≥ 4 25/5	verify_v5.py	out/v5.json
Fixed-121 A/B/C × 3 weightings	see report §9 table	verify_v5.py	out/v5.json
24h day-cluster vs 5-day block	BTC [−13.5,0.0] vs [−13.0,+0.1]	verify_v5.py	out/v5.json
Top-decile gross / cost-adj / fade-adj	+14.0 [+5.4,+22.8] / +7.1 / +1.0 [−10.8,+13.7] p=0.44	cohort_M2_deploy.py	ledger

Claim	Value	Script	Output
Market-state residual + MDE	-1.3 [-15.6,+13.0] / +5.9 [-5.5,+17.7]; MDE 11-16	cohort_M4_residual.py	ledger
Nested incremental IC Behavioral (dir-aligned medians)	-0.0003 recurring vs rest: signed -73/-11, abs 168/131, stretch 308/199, vol 23/19	cohort_P2_freeze_models.py verify_v5.py	out/cohort_P2_models.npz out/v5.json
Winner's-curse OOS	-28.7 bp	oos_persistence.py	ledger

1b. v6 robustness objects

Object	Value	Script	Output
Eligibility leak (train-only recompute)	1,557/1,694 still qualify; 108/121 clean	eligibility_audit.py	out/eligibility_train.parquet
Fixed-121 A/B/C/D × 3 weightings	see report §9	verify_v6.py	out/v6.json
Absolute cost-adjusted p(>0)	0.34 / 0.92 / 0.25	verify_v6.py	out/v6.json
Clean-108 subset B (wallet-day)	+11.1 [4.3,17.5] p<0.001	verify_v6.py	out/v6.json
B moving-block (7d / 5d)	+9.8 p=0.006 / p=0.004	verify_v6.py	out/v6.json
B by coin / coin-balanced / ex-SOL	7.1/17.0/13.6/7.5 · 11.3 · 9.6	verify_v6.py	out/v6.json
Rolling fold ICs (10) / sign-test	mean +0.11, 9/10 pos, p=0.02	verify_v6.py	out/v6.json
Behavioral equal-wallet	signed -81/-9, abs 178/133, stretch 314/196	verify_v6.py	out/v6.json

2. Audit trail

v6 corrections (eligibility leak, absolute-return inference, weighting reframe, portfolio rule, moving-block, coin decomposition, rolling fold inference, behavioral unit, nested-statistic removal): docs/FINAL_AUDIT_V6.md, with supporting tables in docs/ELIGIBILITY_WINDOW_AUDIT.md, docs/FIXED_COHORT_ROBUSTNESS.md, docs/ROLLING_TIME_INFERENCE.md, docs/PORTFOLIO_RULE.md. Prior rounds: docs/FINAL_AUDIT_V5.md, docs/FIXED_121_ESTIMANDS.md, docs/RECURRENCE_TRAIN_VS_FULL.md, docs/CHANGELOG_v2.md, docs/CHANGELOG_v4.md, docs/FINDINGS_LEDGER.md.

3. Reproducibility

Run order (RAM-safe on an 8 GB box; one heavy job at a time; POLARS_MAX_THREADS set to 3 or fewer):

1. build_entries.py -> out/entries/part_*.parquet
2. cohort_K_define.py, cohort_K_price.py -> out/cohort_K_entries.parquet
3. cohort_M_freeze.py -> out/cohort_M_frozen.{txt,parquet}
4. pi_compute.py (heavy, bar-pricing) -> out/eventstudy.json, out/entry_features.parquet
5. verify_v2.py -> out/termstructure_ci.json, out/core_markout_table.md
6. verify_v4.py, verify_v5.py (light) -> out/v4.json, out/v5.json
7. fig_v4.py, fig_v5.py -> figs_v5/*.png
8. pandoc report_v5.md --pdf-engine=xelatex -o report_v5.pdf

Inputs: 5-minute candle closes and cand2 taker fills. No BBO/order-book data is available, so execution-quality markout (fill-to-mid, spread, impact, adverse selection) is not computable and is stated as a limitation.